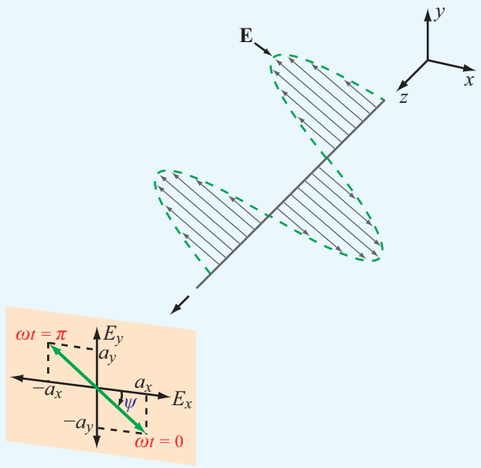



**Figure 7-7** Linearly polarized wave traveling in the +z direction (out of the page).

$$\begin{aligned}\mathbf{E}(z,t) &= \Re\mathfrak{e}\left[\widetilde{\mathbf{E}}(z)\, e^{j\omega t}\right] \\ &= \hat{\mathbf{x}}a\cos(\omega t - kz) + \hat{\mathbf{y}}a\cos(\omega t - kz + \pi/2) \\ &= \hat{\mathbf{x}}a\cos(\omega t - kz) - \hat{\mathbf{y}}a\sin(\omega t - kz). \end{aligned} \tag{7.55b}$$

The corresponding field magnitude and inclination angle are

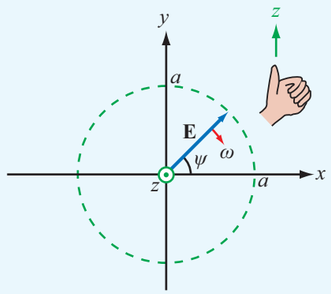
$$\begin{aligned}|\mathbf{E}(z,t)| &= \left[E_x^2(z,t) + E_y^2(z,t)\right]^{1/2} \\ &= \left[a^2\cos^2(\omega t - kz) + a^2\sin^2(\omega t - kz)\right]^{1/2} = a \end{aligned} \tag{7.56a}$$

and

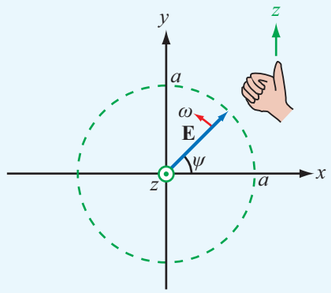
$$\begin{aligned}\psi(z,t) &= \tan^{-1}\left[\frac{E_y(z,t)}{E_x(z,t)}\right] \\ &= \tan^{-1}\left[\frac{-a\sin(\omega t - kz)}{a\cos(\omega t - kz)}\right] = -(\omega t - kz). \end{aligned} \tag{7.56b}$$

We observe that the magnitude of $\mathbf{E}$ is independent of both $z$ and $t$, whereas $\psi$ depends on both variables. These functional dependencies are the converse of those for the linear polarization case.

At $z = 0$, Eq. (7.56b) gives $\psi = -\omega t$; the negative sign implies that the inclination angle decreases as time increases. As illustrated in **Fig. 7-8(a)**, the tip of $\mathbf{E}(t)$ traces a circle in the $x$–$y$ plane and rotates in a clockwise direction as a function of time (when viewing the wave approaching). Such a wave is called ***left-hand circularly polarized*** because when the thumb of the left hand points along the direction of propagation (the $z$ direction in this case) the other four fingers point in the direction of rotation of $\mathbf{E}$.




(a) LHC polarization




(b) RHC polarization

**Figure 7-8** Circularly polarized plane waves propagating in the +z direction (out of the page).

### Right-Hand Circular (RHC) Polarization

For $a_x = a_y = a$ and $\delta = -\pi/2$, we have

$$|\mathbf{E}(z,t)| = a, \qquad \psi = (\omega t - kz). \tag{7.57}$$

The trace of $\mathbf{E}(0,t)$ as a function of $t$ is shown in **Fig. 7-8(b)**. For RHC polarization, the fingers of the right hand point in the direction of rotation of $\mathbf{E}$ when the thumb is along the propagation direction. **Figure 7-9** depicts a right-hand circularly polarized wave radiated by a helical antenna.





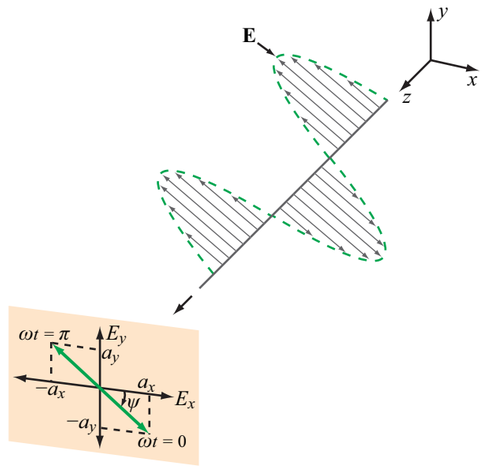



图 7-7 沿 +z 方向传播的线偏振波（纸外）。

$$\begin{aligned}\mathbf{E}(z,t) &= \Re\mathfrak{e}\left[\widetilde{\mathbf{E}}(z)\, e^{j\omega t}\right] \\ &= \hat{\mathbf{x}}a\cos(\omega t - kz) + \hat{\mathbf{y}}a\cos(\omega t - kz + \pi/2) \\ &= \hat{\mathbf{x}}a\cos(\omega t - kz) - \hat{\mathbf{y}}a\sin(\omega t - kz). \end{aligned} \tag{7.55b}$$

相应的场强和倾角为

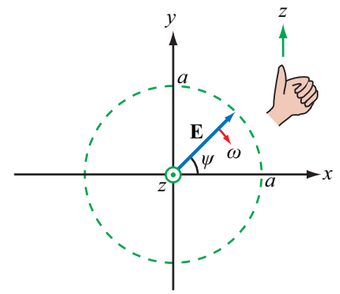
$$\begin{aligned}|\mathbf{E}(z,t)| &= \left[E_x^2(z,t) + E_y^2(z,t)\right]^{1/2} \\ &= \left[a^2\cos^2(\omega t - kz) + a^2\sin^2(\omega t - kz)\right]^{1/2} = a \end{aligned} \tag{7.56a}$$

和

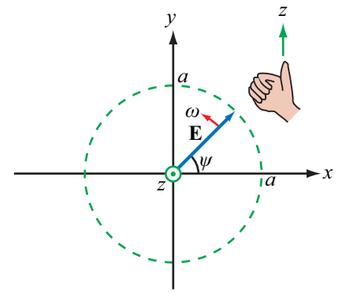
$$\begin{aligned}\psi(z,t) &= \tan^{-1}\left[\frac{E_y(z,t)}{E_x(z,t)}\right] \\ &= \tan^{-1}\left[\frac{-a\sin(\omega t - kz)}{a\cos(\omega t - kz)}\right] = -(\omega t - kz). \end{aligned} \tag{7.56b}$$

我们观察到 E 的大小与 z 和 t 无关，而 ψ 取决于这两个变量。这些函数依赖性与线性偏振情况相反。

在 z = 0 处，方程(7.56b) 给出 ψ = −ωt；负号表示倾角随着时间的增加而减小。如图7-8(a)所示，E(t)的尖端在




(a) LHC polarization




(b) RHC polarization

图 7-8 沿 +z 方向（页面外）传播的圆偏振平面波。

x-y 平面并作为时间的函数沿顺时针方向旋转（当观察波接近时）。这种波称为左手圆偏振波，因为当左手拇指指向传播方向（本例中为 z 方向）时，其他四个手指指向 E 的旋转方向。

右旋圆 (RHC) 偏振

对于 $a_x = a_y = a$ 和 $\delta = -\pi/2$，我们有

$$|\mathbf{E}(z,t)| = a, \qquad \psi = (\omega t - kz). \tag{7.57}$$

E(0,t) 作为 t 函数的轨迹如图 7-8(b) 所示。对于 RHC 偏振，当拇指沿传播方向时，右手手指指向 E 的旋转方向。图 7-9 描绘了螺旋天线辐射的右旋圆极化波。

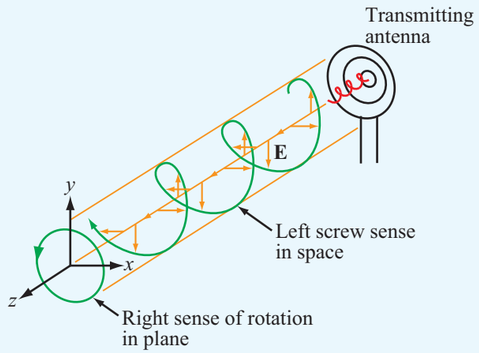



**Figure 7-9** Right-hand circularly polarized wave radiated by a helical antenna.

► ***Polarization handedness*** is defined in terms of the rotation of **E** as a function of time in a fixed plane orthogonal to the direction of propagation, which is opposite the direction of rotation of **E** as a function of distance at a fixed point in time. ◄

## Example 7-2: RHC-Polarized Wave

An RHC-polarized plane wave with electric field magnitude of 3 (mV/m) is traveling in the $+y$ direction in a dielectric medium with $\varepsilon = 4\varepsilon_0$, $\mu = \mu_0$, and $\sigma = 0$. If the frequency is 100 MHz, obtain expressions for $\mathbf{E}(y,t)$ and $\mathbf{H}(y,t)$.

**Solution:** Since the wave is traveling in the $+y$ direction, its field must have components along the $x$ and $z$ directions. The rotation of $\mathbf{E}(y,t)$ is depicted in **Fig. 7-10**, where $\hat{\mathbf{y}}$ is out of the page. By comparison with the RHC-polarized wave shown in **Fig. 7-8(b)**, we assign the $z$ component of $\widetilde{\mathbf{E}}(y)$ a phase angle of zero and the $x$ component a phase shift of $\delta = -\pi/2$. Both components have magnitudes of $a = 3$ (mV/m). Hence,

$$\widetilde{\mathbf{E}}(y) = \hat{\mathbf{x}}\widetilde{E}_x + \hat{\mathbf{z}}\widetilde{E}_z = \hat{\mathbf{x}}ae^{-j\pi/2}e^{-jky} + \hat{\mathbf{z}}ae^{-jky}$$
$$= (-\hat{\mathbf{x}}j + \hat{\mathbf{z}})3e^{-jky} \qquad \text{(mV/m)},$$

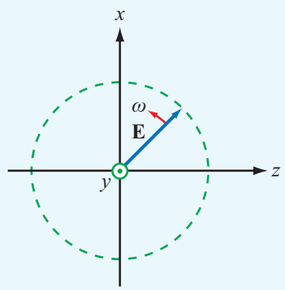



**Figure 7-10** Right-hand circularly polarized wave of Example 7-2.

and application of Eq. (7.39a) gives

$$\widetilde{\mathbf{H}}(y) = \frac{1}{\eta}\hat{\mathbf{y}} \times \widetilde{\mathbf{E}}(y) = \frac{1}{\eta}\hat{\mathbf{y}} \times (-\hat{\mathbf{x}}j + \hat{\mathbf{z}})3e^{-jky}$$
$$= \frac{3}{\eta}(\hat{\mathbf{z}}j + \hat{\mathbf{x}})e^{-jky} \qquad \text{(mA/m)}.$$

With $\omega = 2\pi f = 2\pi \times 10^8$ (rad/s), the wavenumber $k$ is

$$k = \frac{\omega\sqrt{\varepsilon_r}}{c} = \frac{2\pi \times 10^8\sqrt{4}}{3 \times 10^8} = \frac{4}{3}\pi \qquad \text{(rad/m)},$$

and the intrinsic impedance $\eta$ is

$$\eta = \frac{\eta_0}{\sqrt{\varepsilon_r}} \approx \frac{120\pi}{\sqrt{4}} = 60\pi \qquad (\Omega).$$

The instantaneous fields $\mathbf{E}(y,t)$ and $\mathbf{H}(y,t)$ are

$$\mathbf{E}(y,t) = \Re\mathfrak{e}\left[\widetilde{\mathbf{E}}(y)\, e^{j\omega t}\right]$$
$$= \Re\mathfrak{e}\left[(-\hat{\mathbf{x}}j + \hat{\mathbf{z}})3e^{-jky}e^{j\omega t}\right]$$
$$= 3[\hat{\mathbf{x}}\sin(\omega t - ky) + \hat{\mathbf{z}}\cos(\omega t - ky)] \qquad \text{(mV/m)}$$

and

$$\mathbf{H}(y,t) = \Re\mathfrak{e}\left[\widetilde{\mathbf{H}}(y)\, e^{j\omega t}\right]$$
$$= \Re\mathfrak{e}\left[\frac{3}{\eta}(\hat{\mathbf{z}}j + \hat{\mathbf{x}})e^{-jky}e^{j\omega t}\right]$$
$$= \frac{1}{20\pi}[\hat{\mathbf{x}}\cos(\omega t - ky) - \hat{\mathbf{z}}\sin(\omega t - ky)] \qquad \text{(mA/m)}.$$





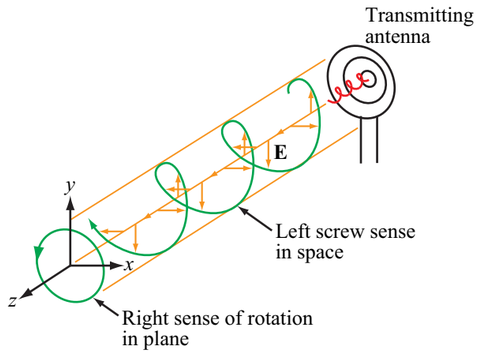



图 7-9 螺旋天线辐射的右旋圆极化波。

► 偏振旋向性是根据 E 在与传播方向正交的固定平面上随时间变化的旋转来定义的，这与 E 在固定时间点随距离变化的旋转方向相反。 ◄

例 7-2：RHC 偏振波

电场强度为 3 (mV/m) 的 RHC 极化平面波在 $\varepsilon = 4\varepsilon_0$、$\mu = \mu_0$ 和 $\sigma = 0$ 的介电介质中沿 +y 方向传播。如果频率为 100 MHz，则获得 E(y,t) 和 H(y,t) 的表达式。

解：由于波沿 +y 方向传播，因此其场必须具有沿 x 和 z 方向的分量。 E(y,t)的旋转如图7-10所示，其中ŷ超出了页面。通过与图7-8(b)所示的RHC偏振波进行比较，我们将 $\widetilde{E}$(y)的z分量指定为0相位角，将x分量指定为$\delta = -\pi/2$的相移。两个分量的幅度均为 = 3 (mV/m)。因此，

$$\widetilde{\mathbf{E}}(y) = \hat{\mathbf{x}}\widetilde{E}_x + \hat{\mathbf{z}}\widetilde{E}_z = \hat{\mathbf{x}}ae^{-j\pi/2}e^{-jky} + \hat{\mathbf{z}}ae^{-jky}$$
$$= (-\hat{\mathbf{x}}j + \hat{\mathbf{z}})3e^{-jky} \qquad \text{(mV/m)},$$

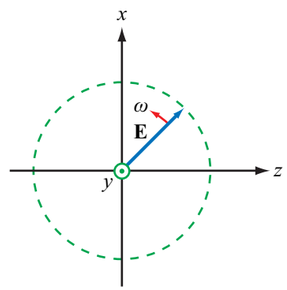



图7-10 例7-2的右旋圆偏振波。

和方程的应用。 (7.39a) 给出

$$\widetilde{\mathbf{H}}(y) = \frac{1}{\eta}\hat{\mathbf{y}} \times \widetilde{\mathbf{E}}(y) = \frac{1}{\eta}\hat{\mathbf{y}} \times (-\hat{\mathbf{x}}j + \hat{\mathbf{z}})3e^{-jky}$$
$$= \frac{3}{\eta}(\hat{\mathbf{z}}j + \hat{\mathbf{x}})e^{-jky} \qquad \text{(mA/m)}.$$

对于 $\omega = 2\pi f = 2\pi \times 10^8$ (rad/s)，波数 k 为

$$k = \frac{\omega\sqrt{\varepsilon_r}}{c} = \frac{2\pi \times 10^8\sqrt{4}}{3 \times 10^8} = \frac{4}{3}\pi \qquad \text{(rad/m)},$$

固有阻抗 η 为

$$\eta = \frac{\eta_0}{\sqrt{\varepsilon_r}} \approx \frac{120\pi}{\sqrt{4}} = 60\pi \qquad (\Omega).$$

瞬时场 E(y,t) 和 H(y,t) 为

$$\mathbf{E}(y,t) = \Re\mathfrak{e}\left[\widetilde{\mathbf{E}}(y)\, e^{j\omega t}\right]$$
$$= \Re\mathfrak{e}\left[(-\hat{\mathbf{x}}j + \hat{\mathbf{z}})3e^{-jky}e^{j\omega t}\right]$$
$$= 3[\hat{\mathbf{x}}\sin(\omega t - ky) + \hat{\mathbf{z}}\cos(\omega t - ky)] \qquad \text{(mV/m)}$$

和

$$\mathbf{H}(y,t) = \Re\mathfrak{e}\left[\widetilde{\mathbf{H}}(y)\, e^{j\omega t}\right]$$
$$= \Re\mathfrak{e}\left[\frac{3}{\eta}(\hat{\mathbf{z}}j + \hat{\mathbf{x}})e^{-jky}e^{j\omega t}\right]$$
$$= \frac{1}{20\pi}[\hat{\mathbf{x}}\cos(\omega t - ky) - \hat{\mathbf{z}}\sin(\omega t - ky)] \qquad \text{(mA/m)}.$$

## Technology Brief 13: RFID Systems

In 1973, two separate patents were issued in the United States for Radio Frequency Identification (RFID) concepts. The first, granted to Mario Cardullo, was for an ***active RFID tag*** with rewritable memory. An active tag has a power source (such as a battery) of its own, whereas a ***passive RFID tag*** does not. The second patent was granted to Charles Walton, who proposed the use of a passive tag for keyless entry (unlocking a door without a key). Shortly thereafter, a passive RFID tag was developed for tracking cattle (**Fig. TF13-1**), and the technology rapidly expanded into many commercial enterprises—from tracking vehicles and consumer products to supply chain management and automobile anti-theft systems.

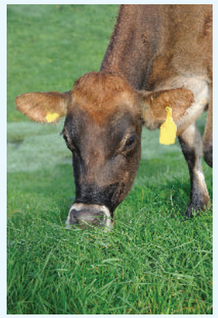


**Figure TF13-1** Passive RFID tags were developed in the 1970s for tracking cows.

### RFID System Overview

In an RFID system, communication occurs between a ***reader***—which actually is a ***transceiver***—and a ***tag*** (**Fig. TF13-2**). When *interrogated* by the reader, a tag responds with information about its identity, as well as other relevant information depending on the specific application.

► Passive RFID systems are limited to short ***read ranges*** (between reader and tag) on the order of 30 cm to 3 m, depending on the system's frequency band (as noted in **Table TT13-1**). ◄

► The tag is, in essence, a ***transponder*** commanded by the reader. ◄

The functionality and associated capabilities of the RFID tag depend on two important attributes: (a) whether the tag is of the active or passive type and (b) the tag's operating frequency. Usually the RFID tag remains dormant (*asleep*) until activated by an electromagnetic signal radiated by the reader's antenna. The magnetic field of the EM signal induces a current in the coil contained in the tag's circuit (**Fig. TF13-3**). For a passive tag, the induced current has to be sufficient to generate the power necessary to activate the chip as well as to transmit the response to the reader.

The obvious advantage of active RFID systems is that they can operate over greater distances and do not require reception of a signal from the reader's antenna to get activated. However, active tags are significantly more expensive to fabricate than their passive cousins.

### RFID Frequency Bands

**Table TT13-1** provides a comparison among the four frequency bands commonly used for RFID systems. Generally speaking, the higher-frequency tags can operate over longer read ranges and can carry higher data rates, but they are more expensive to fabricate.





## 技术简介 13：RFID 系统

1973 年，美国就射频识别 (RFID) 概念颁发了两项单独的专利。第一个授予 Mario Cardullo，用于具有可重写内存的有源 RFID 标签。有源标签有自己的电源（例如电池），而无源 RFID 标签则没有。第二项专利授予查尔斯·沃尔顿（Charles Walton），他提议使用无源标签进行无钥匙进入（无需钥匙即可解锁门）。此后不久，用于跟踪牛的无源RFID标签被开发出来（图TF13-1），并且该技术迅速扩展到许多商业企业——从跟踪车辆和消费品到供应链管理和汽车防盗系统。

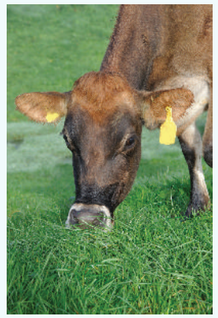


图 TF13-1 无源 RFID 标签于 20 世纪 70 年代开发，用于跟踪奶牛。

### RFID系统概述

在 RFID 系统中，通信发生在读取器（实际上是收发器）和标签之间（图 TF13-2）。当阅读器询问时，标签会响应有关其身份的信息，以及取决于特定应用的其他相关信息。

► 无源 RFID 系统仅限于 30 cm 至 3 m 左右的短读取范围（读取器和标签之间），具体取决于系统的频段（如表 TT13-1 中所述）。 ◄

► 标签本质上是一个由阅读器控制的应答器。 ◄

RFID 标签的功能和相关能力取决于两个重要属性：(a) 标签是有源类型还是无源类型，以及 (b) 标签的工作频率。通常，RFID 标签保持休眠（睡眠）状态，直到被读取器天线辐射的电磁信号激活。 EM 信号的磁场会在标签电路中的线圈中感应出电流（图 TF13-3）。对于无源标签，感应电流必须足以产生激活芯片以及将响应传输到读取器所需的功率。

有源 RFID 系统的明显优势是它们可以在更远的距离上运行，并且不需要接收来自阅读器天线的信号即可激活。然而，有源标签的制造成本明显高于无源标签。

### RFID 频段

表 TT13-1 提供了 RFID 系统常用的四个频段之间的比较。一般来说，高频标签可以在更长的读取范围内运行，并且可以承载更高的数据速率，但它们的制造成本更高。

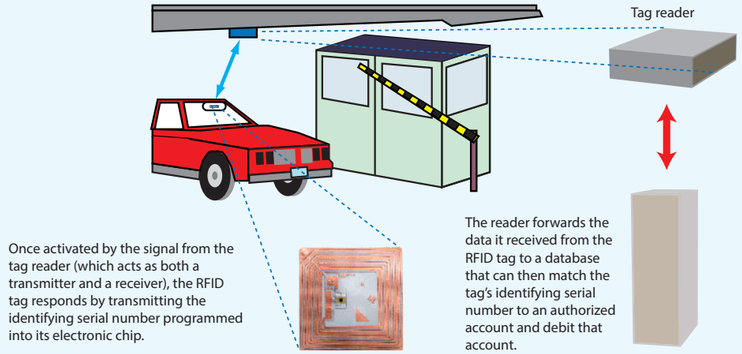



**Figure TF13-2** How an RFID system works is illustrated through this EZ-Pass example. The image of the UHF RFID tag is courtesy of Cary Wolinsky/Cavan Images/Alamy Stock Photo.

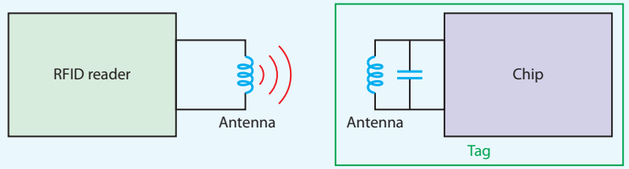



**Figure TF13-3** Simplified diagram for how the RFID reader communicates with the tag. At the two lower carrier frequencies commonly used for RFID communication (125 kHz and 13.56 MHz), coil inductors act as magnetic antennas. In systems designed to operate at higher frequencies (900 MHz and 2.54 GHz), dipole antennas are used instead.

**Table TT13-1 Comparison of RFID frequency bands.**

| Band | LF | HF | UHF | Microwave |
|---|---|---|---|---|
| RFID frequency | 125–134 kHz | 13.56 MHz | 865–956 MHz | 2.45 GHz |
| Read range | $\leq$ 0.5 m | $\leq$ 1.5 m | $\leq$ 5 m | $\leq$ 10 m |
| Data rate | 1 kbit/s | 25 kbit/s | 30 kbit/s | 100 kbit/s |
| Typical applications | • Animal ID<br>• Automobile key/antitheft<br>• Access control | • Smart cards<br>• Article surveillance<br>• Airline baggage tracking<br>• Library book tracking | • Supply chain management<br>• Logistics | • Vehicle toll collection<br>• Railroad car monitoring |



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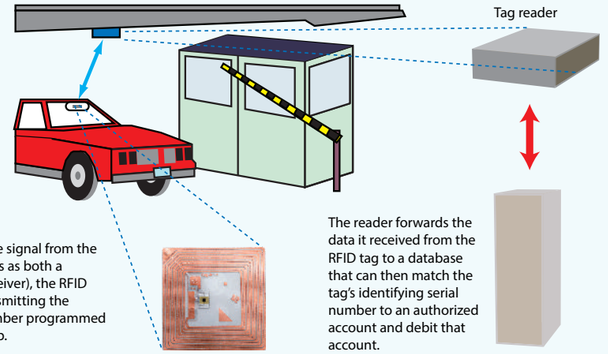



图 TF13-2 通过此 EZ-Pass 示例说明了 RFID 系统的工作原理。 UHF RFID 标签的图像由 Cary Wolinsky/Cavan Images/Alamy Stock Photo 提供。

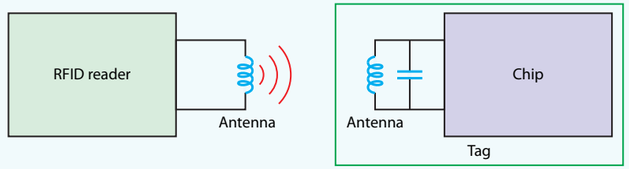



图 TF13-3 RFID 阅读器如何与标签通信的简化图。在 RFID 通信常用的两个较低载波频率（125 kHz 和 13.56 MHz）下，线圈电感器充当磁性天线。在设计为在较高频率（900 MHz 和 2.54 GHz）下运行的系统中，改为使用偶极天线。

表TT13-1 RFID频段比较。

| Band | LF | HF | UHF | Microwave |
|---|---|---|---|---|
| RFID frequency | 125–134 kHz | 13.56 MHz | 865–956 MHz | 2.45 GHz |
| Read range | $\leq$ 0.5 m | $\leq$ 1.5 m | $\leq$ 5 m | $\leq$ 10 m |
| Data rate | 1 kbit/s | 25 kbit/s | 30 kbit/s | 100 kbit/s |
| Typical applications | • Animal ID<br>• Automobile key/antitheft<br>• Access control | • Smart cards<br>• Article surveillance<br>• Airline baggage tracking<br>• Library book tracking | • Supply chain management<br>• Logistics | • Vehicle toll collection<br>• Railroad car monitoring |

### 7-3.3 Elliptical Polarization

Plane waves that are not linearly or circularly polarized are elliptically polarized. That is, the tip of $\mathbf{E}(z,t)$ traces an ellipse in the plane perpendicular to the direction of propagation. The shape of the ellipse and the field's handedness (left-hand or right-hand) are determined by the values of the ratio $(a_y/a_x)$ and the phase difference $\delta$.

The polarization ellipse shown in **Fig. 7-11** has its major axis with length $a_\xi$ along the $\xi$ direction and its minor axis with length $a_\eta$ along the $\eta$ direction. The ***rotation angle*** $\gamma$ is defined as the angle between the major axis of the ellipse and a reference direction, which is chosen here to be the $x$ axis, with $\gamma$ being bounded within the range $-\pi/2 \le \gamma \le \pi/2$. The shape of the ellipse and its handedness are characterized by the ***ellipticity angle*** $\chi$, which is defined as

$$\tan\chi = \pm\frac{a_\eta}{a_\xi} = \pm\frac{1}{R}, \tag{7.58}$$

with the plus sign corresponding to left-handed rotation and the minus sign corresponding to right-handed rotation. The limits for $\chi$ are $-\pi/4 \le \chi \le \pi/4$. The quantity $R = a_\xi/a_\eta$ is called the ***axial ratio*** of the polarization ellipse, and it varies between 1 for circular polarization and $\infty$ for linear polarization. The polarization angles $\gamma$ and $\chi$ are related to the wave parameters $a_x$, $a_y$, and $\delta$ by*

$$\tan 2\gamma = (\tan 2\psi_0)\cos\delta \quad (-\pi/2 \le \gamma \le \pi/2), \tag{7.59a}$$
$$\sin 2\chi = (\sin 2\psi_0)\sin\delta \quad (-\pi/4 \le \chi \le \pi/4), \tag{7.59b}$$

*From M. Born and E. Wolf, *Principles of Optics*, New York: Macmillan, 1965, p. 27.

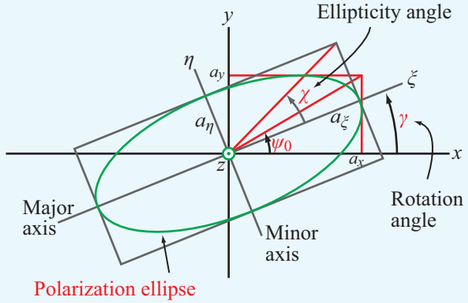



**Figure 7-11** Polarization ellipse in the $x$–$y$ plane with the wave traveling in the $z$ direction (out of the page).

where $\psi_0$ is an ***auxiliary angle*** defined by

$$\tan\psi_0 = \frac{a_y}{a_x} \qquad \left(0 \le \psi_0 \le \frac{\pi}{2}\right). \tag{7.60}$$

Sketches of the polarization ellipse are shown in **Fig. 7-12** for various combinations of the angles $(\gamma,\chi)$. The ellipse reduces to a circle for $\chi = \pm 45°$ and to a line for $\chi = 0$.

► Positive values of $\chi$, corresponding to $\sin\delta > 0$, are associated with left-handed rotation, and negative values of $\chi$, corresponding to $\sin\delta < 0$, are associated with right-handed rotation. ◄

Since the magnitudes $a_x$ and $a_y$ are, by definition, nonnegative numbers, the ratio $a_y/a_x$ may vary between zero for an $x$-polarized linear polarization and $\infty$ for a $y$-polarized linear polarization. Consequently, the angle $\psi_0$ is limited to the range $0 \le \psi_0 \le 90°$. Application of Eq. (7.59a) leads to two possible solutions for the value of $\gamma$, both of which fall within the defined range from $-\pi/2$ to $\pi/2$. The correct choice is governed by the following rule:

$$\gamma > 0 \text{ if } \cos\delta > 0,$$
$$\gamma < 0 \text{ if } \cos\delta < 0.$$

► In summary, the sign of the rotation angle $\gamma$ is the same as the sign of $\cos\delta$, and the sign of the ellipticity angle $\chi$ is the same as the sign of $\sin\delta$. ◄

**Example 7-3: Polarization State**

Determine the polarization state of a plane wave with electric field

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}\,3\cos(\omega t - kz + 30°) - \hat{\mathbf{y}}\,4\sin(\omega t - kz + 45°) \quad \text{(mV/m)}.$$

**Solution:** We begin by converting the second term to a cosine reference:

$$\mathbf{E} = \hat{\mathbf{x}}\,3\cos(\omega t - kz + 30°) - \hat{\mathbf{y}}\,4\cos(\omega t - kz + 45° - 90°)$$
$$= \hat{\mathbf{x}}\,3\cos(\omega t - kz + 30°) - \hat{\mathbf{y}}\,4\cos(\omega t - kz - 45°).$$

The corresponding field phasor $\tilde{\mathbf{E}}(z)$ is

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}}\,3e^{-jkz}e^{j30°} - \hat{\mathbf{y}}\,4e^{-jkz}e^{-j45°}$$
$$= \hat{\mathbf{x}}\,3e^{-jkz}e^{j30°} + \hat{\mathbf{y}}\,4e^{-jkz}e^{-j45°}e^{j180°}$$
$$= \hat{\mathbf{x}}\,3e^{-jkz}e^{j30°} + \hat{\mathbf{y}}\,4e^{-jkz}e^{j135°},$$





### 7-3.3 椭圆偏振

非线偏振或圆偏振的平面波为椭圆偏振。即，E(z,t) 的尖端在垂直于传播方向的平面中描绘出椭圆形。椭圆的形状和场的旋向（左旋或右旋）由比率 $(a_y/a_x)$ 和相位差 δ 的值决定。

图7-11所示的偏振椭圆的长轴沿ξ方向长度为$a_\xi$，短轴沿η方向长度为$a_\eta$。旋转角 γ 定义为椭圆长轴与参考方向（此处选择为 x 轴）之间的角度，其中 γ 被限制在范围 $-\pi/2 \le \gamma \le \pi/2$ 内。椭圆的形状及其旋向由椭圆度角 χ 来表征，其定义为

$$\tan\chi = \pm\frac{a_\eta}{a_\xi} = \pm\frac{1}{R}, \tag{7.58}$$

其中加号对应于左手旋转，减号对应于右手旋转。 χ 的限制为 $-\pi/4 \le \chi \le \pi/4$。 $R = a_\xi/a_\eta$ 称为偏振椭圆的轴比，它在圆偏振的 1 和线偏振的 ∞ 之间变化。偏振角γ和χ与波参数$a_x$、$a_y$和δ通过*相关

$$\tan 2\gamma = (\tan 2\psi_0)\cos\delta \quad (-\pi/2 \le \gamma \le \pi/2), \tag{7.59a}$$
$$\sin 2\chi = (\sin 2\psi_0)\sin\delta \quad (-\pi/4 \le \chi \le \pi/4), \tag{7.59b}$$

*来自 M. Born 和 E. Wolf，《光学原理》，纽约：Macmillan，1965 年，第 14 页。 27.

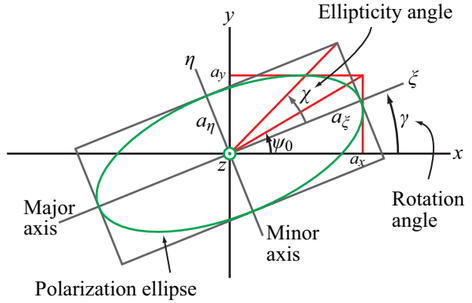



图 7-11 x-y 平面中的偏振椭圆，波沿 z 方向传播（页面外）。

其中 $\psi_0$ 是由下式定义的辅助角

$$\tan\psi_0 = \frac{a_y}{a_x} \qquad \left(0 \le \psi_0 \le \frac{\pi}{2}\right). \tag{7.60}$$

各种角度组合（γ、χ）的偏振椭圆草图如图 7-12 所示。对于 $\chi = \pm 45°$，椭圆缩小为圆形；对于 $\chi = 0$，椭圆缩小为直线。

► χ 的正值对应于 $\sin\delta > 0$，与左手旋转相关联，而 χ 的负值对应于 $\sin\delta < 0$，与右手旋转相关联。◄

由于根据定义，幅度 $a_x$ 和 $a_y$ 是非负数，因此比率 $a_y/a_x$ 可以在 x 偏振线性偏振的零和 y 偏振线性偏振的 ∞ 之间变化。因此，角度$\psi_0$被限制在$0 \le \psi_0 \le$至90°的范围内。方程的应用(7.59a) 得出 γ 值的两个可能的解，它们都落在从 $-\pi/2$ 到 $\pi/2$ 的定义范围内。正确的选择遵循以下规则：

$$\gamma > 0 \text{ if } \cos\delta > 0,$$
$$\gamma < 0 \text{ if } \cos\delta < 0.$$

► 综上所述，旋转角γ的符号与cosδ的符号相同，椭圆角χ的符号与sinδ的符号相同。◄

例 7-3： 偏振态

确定电场中平面波的偏振态

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}\,3\cos(\omega t - kz + 30°) - \hat{\mathbf{y}}\,4\sin(\omega t - kz + 45°) \quad \text{(mV/m)}.$$

解决方案：我们首先将第二项转换为余弦参考：

$$\mathbf{E} = \hat{\mathbf{x}}\,3\cos(\omega t - kz + 30°) - \hat{\mathbf{y}}\,4\cos(\omega t - kz + 45° - 90°)$$
$$= \hat{\mathbf{x}}\,3\cos(\omega t - kz + 30°) - \hat{\mathbf{y}}\,4\cos(\omega t - kz - 45°).$$

相应的场相量 $\tilde{}$E(z) 为

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}}\,3e^{-jkz}e^{j30°} - \hat{\mathbf{y}}\,4e^{-jkz}e^{-j45°}$$
$$= \hat{\mathbf{x}}\,3e^{-jkz}e^{j30°} + \hat{\mathbf{y}}\,4e^{-jkz}e^{-j45°}e^{j180°}$$
$$= \hat{\mathbf{x}}\,3e^{-jkz}e^{j30°} + \hat{\mathbf{y}}\,4e^{-jkz}e^{j135°},$$